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Modifying $\text{Li}_2\text{B}_{12}\text{H}_{12}$ to become a highly electrochemically stable solid electrolyte†Thi Phuong Thao Nguyen, ^a Wanli Gao, ^a Terry D. Humphries, ^a
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Metal hydridoborates are promising electrolyte materials for next generation batteries. However, most hydridoborates show low ionic conductivity at room temperature and relatively poor electrochemical stability towards high-voltage cathodes. In this work, modifications to the $\text{Li}_2\text{B}_{12}\text{H}_{12}$ solid electrolyte are undertaken by ball-milling, partial thermal decomposition and hydration. Impacts of morphological and structural changes on their ionic conductivity and electrochemical stability are analysed and discussed. A solid-state electrolyte synthesised by partial thermal decomposition of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ at 380 °C is reported to exhibit remarkable oxidative stability, greater than 6 V (vs. Li^+/Li), and an increase by nearly 2 orders of magnitude in Li^+ conductivity at 30 °C compared to the pristine material. Herein, partial thermal decomposition is proposed as a new approach to improve the electrochemical properties of metal hydridoborate-based solid electrolytes.

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Introduction

All-solid-state batteries (ASSBs) are proposed as next generation batteries, and have the potential to provide high energy density while overcoming the safety and cost issues of conventional Li-ion batteries.^{1,2} The key technology in ASSBs is the utilisation of solid-state electrolytes (SSEs) instead of volatile and flammable organic liquid electrolytes. To serve as a SSE, a material should not only have high ionic conductivity and electrochemical stability, but also good mechanical properties for battery fabrication and cycling performance.

In 2007, superionic conduction in lithium borohydride (LiBH_4) was reported by Matsuo *et al.*³ leading to vast research efforts into understanding the ionic conductivity and electrochemical properties of metal hydridoborates as a SSE material genre.⁴ Metal hydridoborates are metal salts of complex anions composed of B bonding with H. The B–H clusters in these compounds can exhibit reorientational dynamics in the solid-state, resulting in a polymorphic phase transition from an ordered crystallographic structure at low temperature to a disordered crystal structure at high temperature.^{2,5,6} It is typical that the disordered crystallographic phase enables much greater cationic conductivity, thus providing promising solid-

state electrolytes. In terms of electrochemical stability, metal hydridoborates are excellent reducing agents, so they are naturally more compatible with metallic negative electrodes and capable of forming a stable and reversible anode–SSE interface.^{2,7} However, their electrochemical stability against high-voltage cathodes remains a concern because they may be readily oxidised. Metal hydridoborates are highly deformable salts and lighter in molecular weight compared with oxide and sulfide SSEs, allowing good interfacial contact with simple uniaxial pressing, providing the possibility to develop light-weight devices.⁷

Recently, studies in the field of hydridoborates have focussed on metal salts of large poly-atomic anions, such as lithium *closo*-dodecahydridodecaborate ($\text{Li}_2\text{B}_{12}\text{H}_{12}$), due to the weakly coordinating nature between cations and the bulky, low charge density anion. $\text{Li}_2\text{B}_{12}\text{H}_{12}$ has a cubic $P\bar{a}3$ crystal structure type at room temperature (RT) and presents a polymorphic phase transition at ≈ 355 °C.^{8–10} Compared with LiBH_4 , $\text{Li}_2\text{B}_{12}\text{H}_{12}$ is more stable (the oxidative stability limit of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ and LiBH_4 is +3.4 and +2.04 V vs. Li^+/Li , respectively)^{11,12} since it is an intermediate decomposition product of LiBH_4 .^{13,14} The formation of $\text{Li}_2\text{B}_{12}\text{H}_{12}$, due to the oxidation of LiBH_4 at the positive electrode in a $\text{TiS}_2|\text{LiBH}_4|\text{Li}$ ASSB, has been proposed to create a stable SSE–cathode interfacial layer, enabling durable battery cycling.¹⁵ Intense efforts have been made to bring the hydridoborate polymorphic phase transition down near RT (which would theoretically enable RT superionic conductivity), as well as further efforts to widen the working potential window of hydridoborates. These studies have included chemical modifications of B–H bonds, nanoconfinement into porous scaffolds, cationic mixtures, anionic mixtures, and mechanochemical

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modification.^{6,11,16–18} For chemical modification, the substitution of one B atom in the $B_{12}H_{12}^{2-}$ anion cage with a C atom yields a monovalent carbaborate anion ($CB_{11}H_{12}^-$) that has a lower temperature polymorphic phase transition at ≈ 130 °C and thus, improves Li^+ conductivity at lower temperatures (≥ 0.15 S cm^{-1} at 130 °C).¹⁹ $CB_{11}H_{12}^-$ also shows a higher oxidative stability limit compared with $B_{12}H_{12}^{2-}$ ($> +2.4$ V compared with $+1.4$ V vs. the saturated calomel electrode (SCE)).⁶ However, the synthesis of metal carbaborates is often complicated and expensive, requiring the use of toxic chemicals such as trifluoromethyltrimethylsilane (CF_3SiMe_3) and $B_{10}H_{14}$.²⁰ A cost-effective reaction pathway using common laboratory reagents has been recently proposed but the highest $CB_{11}H_{12}^-$ yield was $\approx 40\%$.²¹ Functionalisation of the $B_{12}H_{12}^{2-}$ anion with halogen (Cl, Br, and I), $-OH$ and $-NH_3$ groups has been shown to result in lower ionic conductivities.^{22–24}

Mechanochemical modification is a straight-forward approach to tune the electrochemical properties of metal hydridoborate-based electrolytes. Ball-milling *closo*-hydridoborates reduces their crystallite size and induces disorder into their crystal structures leading to the formation of ‘high-temperature-disordered-structure-like’ phases at RT, therefore improving ionic conductivity.²⁵ Defects such as atomic vacancies caused by ball-milling can also lead to enhanced ionic conductivity.^{18,25} A recent publication proposed ball-milling with inert oxides as an approach to improve the ionic conductivity and electrochemical stability of $Li_2B_{12}H_{12}$.¹¹ A composite synthesised by ball-milling $Li_2B_{12}H_{12}$ with alumina (Al_2O_3) in a 25% volume ratio for 120 h was reported to exhibit an increase in Li^+ conductivity by 3 orders of magnitude and provide stability up to $+3.8$ V vs. Li^+/Li .¹¹ However, drawbacks of this approach are that aggressive milling conditions (high energy and extremely long duration) are required and the inert oxide additive may result in ‘dead weight’. Besides the above-mentioned methods, there are other modification strategies which show positive outcomes for metal hydridoborate compounds. For example, solvation with neutral molecules such as H_2O , NH_3 , CH_3NH_2 , and THF has been reported to result in higher ionic conductivity of $LiBH_4$,^{26–28} $ZnB_{12}H_{12}$ and $MgB_{12}H_{12}$,²⁹ alkali salts of $B_{11}H_{14}^-$,³⁰ and $Mg(BH_4)_2$.^{31–33}

Based on the current state of the art, two low-cost modification approaches to tune the electrochemical properties of $Li_2B_{12}H_{12}$ have been investigated: disorder by applying external energy and solvation. To introduce disorder within the $Li_2B_{12}H_{12}$ crystal structure, thermal energy was used to partially decompose $Li_2B_{12}H_{12}$ in comparison to results obtained from conventional mechanical ball-milling. The solvation approach was studied by forming a series of hydrated $Li_2B_{12}H_{12}$ samples, taking advantage of the highly hygroscopic nature of the compound.

Experimental

Lithium *closo*-dodecahydrododecaborate tetrahydrate ($Li_2B_{12}H_{12} \cdot 4H_2O$, $>98\%$, Katchem) was used as a starting material. All sample handling was performed in an argon (Ar) glovebox (MBraun $p(H_2O, O_2) < 1$ ppm).

Anhydrous $Li_2B_{12}H_{12}$ (pristine $Li_2B_{12}H_{12}$) was obtained by dehydration of $Li_2B_{12}H_{12} \cdot 4H_2O$ at ≈ 250 °C for 20 h *in vacuo*. Ball-milled $Li_2B_{12}H_{12}$ samples were prepared by unidirectionally milling ≈ 0.5 g of pristine $Li_2B_{12}H_{12}$ for 5–140 h without breaks at a rotation speed of 400 rpm and a ball-to-powder weight ratio of 20 : 1 using an Across International planetary ball mill (PQ-N04). A 50 mL 316 stainless steel milling jar and 8 mm stainless steel balls were used for milling up to 5 h and a 50 mL zirconia jar and 5 mm zirconia balls were used for the longer durations. Both vials were sealed under argon to prevent air exposure.

Partial thermal decomposition of $Li_2B_{12}H_{12}$ was conducted by heating $Li_2B_{12}H_{12} \cdot 4H_2O$ at either 380, 450 and 610 °C *in vacuo* for 1 or 6 h in a stainless-steel reactor, after ramping from room temperature at 4 °C min^{-1} . Samples obtained after thermal decomposition to 380 and 450 °C had a light pink colour, whilst a sample obtained after decomposition to 610 °C was dark brown. The thermally decomposed samples were found to be insoluble in common solvents used for nuclear magnetic resonance (NMR) spectroscopy such as deuterium oxide (D_2O), acetonitrile- d_3 (CD_3CN) and dimethyl sulfoxide- d_6 ($DMSO-d_6$), unlike pristine $Li_2B_{12}H_{12}$, which has high solubility.³⁴ Hereafter, samples synthesised by partial thermal decomposition are referred to as $Li_2B_{12}H_{12-y}$.³⁵ Thermal decomposition at selected temperatures was separately analysed by thermal gravimetric analysis (TGA) with 10–11 mg of $Li_2B_{12}H_{12} \cdot 4H_2O$ in open Al_2O_3 crucibles on a Discovery HP-TGA 750 high pressure thermogravimetric analyser (TA Instruments, balance precision of ± 0.1 μg). The samples were heated from RT to the desired temperatures (380, 450 and 600 °C; $\Delta T/\Delta t = 4$ °C min^{-1}) followed by a 6 h isotherm under 10 mL per min Ar. Before measurements, the samples were placed under vacuum before being purged with Ar to remove any air. Water removal was confirmed by mass loss below 250 °C, whereas partial decomposition of $Li_2B_{12}H_{12}$ was observed above 250 °C by hydrogen loss (see Fig. S1†).³⁵ During TGA, after the initial decomposition, we observed considerably stable mass during isothermal treatment, suggesting that effects of decomposition time could be negligible in small lab-scale synthesis. H_2 loss was measured to be ~ 0.1 , 3.7 and 4.3 mass%, assumed to be ~ 0.3 , 8.4 and 9.8H atoms/ $Li_2B_{12}H_{12}$ molecule for the partial decomposition at 380, 450 and 610 °C.

$Li_2B_{12}H_{12} \cdot xH_2O$ samples with $0 < x < 4$ were synthesised by unidirectionally ball milling mixtures of pristine $Li_2B_{12}H_{12}$ with $Li_2B_{12}H_{12} \cdot 4H_2O$ at molar ratios of 1 : 0.4–1 : 3.2 (total sample weight ≈ 0.5 g) for 10 min at a speed of 400 rpm and a ball-to-powder weight ratio of 20 : 1. Air-tight 50 mL 316 stainless steel jars and 8 mm stainless steel balls were used under an Ar atmosphere. For comparison, both pristine $Li_2B_{12}H_{12}$ and $Li_2B_{12}H_{12} \cdot 4H_2O$ were also individually ball-milled with the same protocols. $Li_2B_{12}H_{12} \cdot 7H_2O$ was obtained by dissolving $Li_2B_{12}H_{12} \cdot 4H_2O$ in an excess amount of Milli-Q water and then slowly evaporating the uncoordinated water at ≈ 0 °C *in vacuo* using Schlenk line techniques. To avoid any loss of coordinated water, $Li_2B_{12}H_{12} \cdot 7H_2O$ was used as is, without ball milling.

Powder X-ray diffraction (XRD) patterns were collected from 5 to 80° 2θ using a Bruker D8 Advance diffractometer (Cu- $K\alpha_{1+2}$

radiation, $\lambda = 1.540562, 1.544398 \text{ \AA}$) equipped with an anti-scatter screen. Samples were packed in either thin-wall 0.7 mm borosilicate capillaries or flat-plate Bruker XRD specimen holders covered with a 2.5 μm airtight X-ray film (Mylar TF-125) to prevent the interference of moisture or air.

Raman spectroscopy measurements were conducted on a WITEC Alpha 300R Raman spectrometer equipped with a CCD detector. Powder samples, sealed in thin-wall 0.7 or 3 mm borosilicate capillaries, were excited with a laser with a wavelength of 532 nm at 15 mW. For the $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ sample synthesised at 610 $^\circ\text{C}$, a laser power of 3 mW was used since the sample was visually burnt at higher laser power. For all samples, a diffraction grating with a groove density of 600 grooves per mm and a blaze wavelength (BLZ) of 500.00 nm was used. Data processing was conducted using WITEC Project 6.2 software.

Fourier transform infrared spectroscopy (FTIR) measurements for the $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples were performed using a Thermo Scientific Nicolet Summit FTIR spectrometer equipped with a diamond attenuated total reflection (ATR) stage in air. All spectra were collected at a resolution of 4 cm^{-1} using 32 scans with minimal air exposure (<1 min).

TGA measurements with the $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ series were performed using a Netzsch STA 449 F3 Jupiter apparatus, heating from 40 to 600 $^\circ\text{C}$ ($\Delta T/\Delta t = 4 \text{ }^\circ\text{C min}^{-1}$) under a 40 mL per min Ar flow. The instrument was calibrated using reference materials (In, Zn, Sn, Bi and CsCl), providing a temperature accuracy of $\pm 0.2 \text{ }^\circ\text{C}$ and balance precision of $\pm 20 \mu\text{g}$. $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ samples ($\sim 10 \text{ mg}$) were sealed in Al crucibles with pierced lids under an Ar atmosphere.

Temperature programmed photographic analysis (TPPA) was conducted on $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$. The sample was sealed in a thin-wall borosilicate capillary ($\Phi = 0.5 \text{ mm}$) under an Ar atmosphere then heated slowly ($\leq 5 \text{ }^\circ\text{C min}^{-1}$) to 200 $^\circ\text{C}$ using a heating block. Transformation of the powder sample was continuously photographed using a HAYEAR HY-5299 digital microscope. To confirm phenomena observed in TPPA, differential scanning calorimetry coupled with TGA (DSC-TGA) was performed with $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$ sealed in an Al crucible under an Ar atmosphere.

Electrochemical impedance spectroscopy (EIS) data were measured from 10^6 to 1 Hz at an amplitude of 50 mV AC using a ZIVE MP10 multichannel electrochemical workstation. The powder samples were pressed at 195 MPa for 1 min to form 8 mm-diameter cylindrical pellets with thicknesses between 0.77 and 1.41 mm and sandwiched between two pieces of 0.1 mm-thick gold foil. The symmetric solid cells were contained in stainless-steel 2-electrode split test cells (MTI Corp.) designed with a spring to apply stack pressure. Measurements were conducted from 30 to 165 $^\circ\text{C}$ with a temperature step of 15 $^\circ\text{C}$ and a soaking time of 30–45 min at each temperature using an Across International forced air convection oven. Ionic conductivity (σ) of the SSEs was calculated from EIS datasets using eqn (1):²²

$$\sigma_{\text{Li}} = L/(R_{\text{total}} \times A) \quad (1)$$

where L is the thickness of the SSE pellet, R_{total} is the resistance determined from the low-frequency x -intercept of the semi-

circle on a Nyquist plot (e.g. Fig. S2†) and A is the contact area of the pellet with the gold electrodes.

Electronic conductivity of the electrolytes was evaluated by the DC method on a ZIVE MP 10 multichannel electrochemical workstation. The symmetric cell system, with two ion-blocking gold electrodes, was utilised. A linear potential scan from an open circuit potential (OCP) to 1 V higher than the OCP was applied to the cells at a scan rate of 5 mV s^{-1} at 80 $^\circ\text{C}$. Current in the nanoampere range (Fig. S3†) corresponding to electrical resistance in the order of $10^9 \Omega$ was observed, suggesting that the SSEs are electronically insulating.

Electrochemical oxidative stability of the SSEs was investigated with linear sweep voltammetry (LSV) on a ZIVE MP 5 multichannel electrochemical workstation using the method reported by Asakura *et al.*¹² with adaptations. Super-P carbon (Sigma Aldrich, $\geq 99\%$) was dehydrated at 115 $^\circ\text{C}$ *in vacuo* for 24 h before use. Sample/carbon composites were made by mixing the samples with the dehydrated carbon at a weight ratio of 75 : 25. First, 40 mg of SSE powder was added into an 8 mm pressing die followed by addition of 4 mg of its carbon composite and then, the powder stack was uniaxially pressed under $\approx 195 \text{ MPa}$ for 1 min to form a 2-layer pellet. The pellet was then sandwiched between a clean 0.38 mm-thick Li counter electrode (99.9%, Sigma-Aldrich) and a 0.1 mm-thick Pt foil of an identical diameter in a stainless steel 2-electrode split test cell (MTI Corp.) with the sample/carbon composite layer (working electrode layer) facing the Pt foil. An Al foil (15 μm thickness) was placed on top of the Pt foil as a current collector. Measurements were conducted at 80 $^\circ\text{C}$, from -0.4 V lower than OCP to +6 V (vs. Li^+/Li) with a scan rate of 0.05 mV s^{-1} .

Galvanostatic charge–discharge (GCD) measurements were conducted with a $\text{Li}|\text{Li}_2\text{B}_{12}\text{H}_{12-y}|\text{Li}$ symmetric cell at 130 $^\circ\text{C}$ on a Metrohm Autolab potentiostat to study the compatibility of the SSE towards Li metal. Critical current density was determined by performing GCD at increasing current densities from 3.0 $\mu\text{A cm}^{-2}$ to 0.41 mA cm^{-2} over $\approx 100 \text{ h}$. The stability of the Li–SSE interface was evaluated by GCD at a constant current density of $\approx 0.17 \text{ mA cm}^{-2}$.

Results and discussion

Structural modifications and ionic conductivity

Ball-milling. Fig. 1 illustrates selected XRD data of ball-milled $\text{Li}_2\text{B}_{12}\text{H}_{12}$ samples with milling durations of 5, 120 and 140 h, focusing on the region of $2\theta = 15\text{--}20^\circ$ (full patterns are provided in Fig. S4†). All samples show Bragg peaks which can be indexed to the space group $Pa\bar{3}$, of which the structure of α - $\text{Li}_2\text{B}_{12}\text{H}_{12}$ has been previously reported.^{8,9} The Bragg peaks shift to lower 2θ angles (higher d -spacing) and broaden as the milling duration increases, indicating smaller crystallite sizes. No impurities and other polymorphic phases of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ (β - and γ -phases)³⁵ are detected. Lattice constants calculated from full XRD patterns of the samples are provided in Table S1.† The sample ball-milled for 140 h shows a notable increase in the lattice cell dimension ($a = 9.60667 \text{ \AA}$) and volume ($V = 886.6 \text{ \AA}^3$), compared with the pristine sample ($a = 9.55764 \text{ \AA}$ and $V = 873.1 \text{ \AA}^3$). The changes in cell parameters could indicate the presence

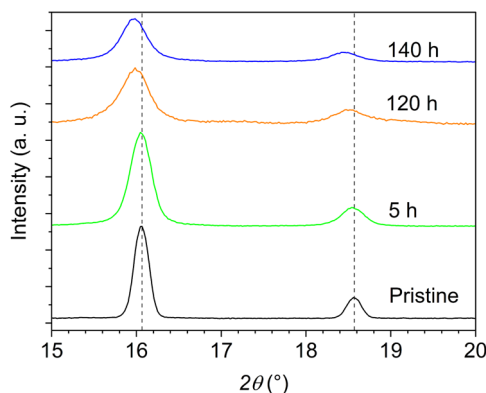


Fig. 1 XRD patterns of ball-milled samples with milling durations of 5, 120 and 140 h in comparison with pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$, measured in borosilicate capillaries.

of tensile strain or disorder in the lattice of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ caused by mechanical ball-milling.

To further understand the effects of ball-milling on the structure of $\text{Li}_2\text{B}_{12}\text{H}_{12}$, Raman spectroscopy measurements were conducted (Fig. 2). Similar to that of the pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$, the ball-milled samples present Raman signals of B–H bending and stretching modes within the bands of 500–1200 and 2400–2600 cm^{-1} , respectively, and signals related to B–B bonding at ≈ 944 and 974 cm^{-1} .^{13,18,36} Bands at 750 and 2500–2600 cm^{-1} are ascribed to symmetric modes, A_g , and the remaining bands to mixed modes, H_g .³⁷ Ball milling for a duration of 5 h causes subtle broadening and downshift of the B–H stretching modes at 2517 and 2547 cm^{-1} to 2514 and 2544 cm^{-1} , respectively. Longer milling durations result in more broadening and substantial shifting of these bands towards lower wavenumbers. In the B–H bending region, only subtle shifting of the H_g mode at 770 cm^{-1} is observed for samples with milling durations of 120 and 140 h. The broadening and shifting of the Raman modes can be attributed to the decrease in the nanocrystallite size but effects of strain cannot be excluded since lattice expansion by ball-milling is evidenced by XRD. The bands between 2500 and 2600 cm^{-1} are ascribed to out-of-phase breathing modes of the icosahedral $\text{B}_{12}\text{H}_{12}^{2-}$, largely controlled by B–H stretching constant.³⁷ Thus, the changes in shape and position of these bands may indicate anharmonic oscillations

of $\text{B}_{12}\text{H}_{12}^{2-}$ cages caused by local distortion of B–H bonds in the ball-milled samples.¹⁸ In addition, since no significant shifting is observed in signals of B–B bonding and B–H bending regions, the band shift may be related to the change in lengths of B–H bonds within the $\text{Li}_2\text{B}_{12}\text{H}_{12}$ lattice after ball-milling.³⁸ Consequently, Raman spectroscopy and XRD results suggest that external energy from mechanical ball-milling induces tensile stresses in the crystal structure of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ leading to the distortion of B–H bonding (likely increasing average B–H bond lengths) and subsequently, expanding the crystal lattice. These changes could have implications for Li^+ conductivity within the crystal structure.

The ionic conductivity of ball-milled samples was measured isothermally at points between 30 and 165 $^{\circ}\text{C}$ using EIS (Fig. 3). Compared with pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$, the samples show an increase of 1–2 orders of magnitude in Li^+ conductivity. The effect of mechanical ball-milling to promote ionic conduction in $\text{Li}_2\text{B}_{12}\text{H}_{12}$ has been previously reported in the literature.^{18,25} There is a good agreement that stresses and defects induced by ball-milling in the lattice may lead to improved conductivity. Here, lattice expansion from the tensile strain may allow for better Li^+ diffusion pathways and enhance anionic reorientational dynamics leading to enhanced cation mobility *via* the

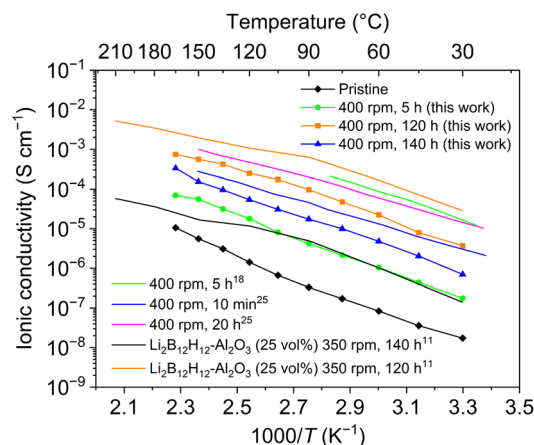


Fig. 3 Ionic conductivity of ball-milled samples in this work (lines with filled symbols, measured isothermally by EIS) and the literature data (lines).^{11,18,25}

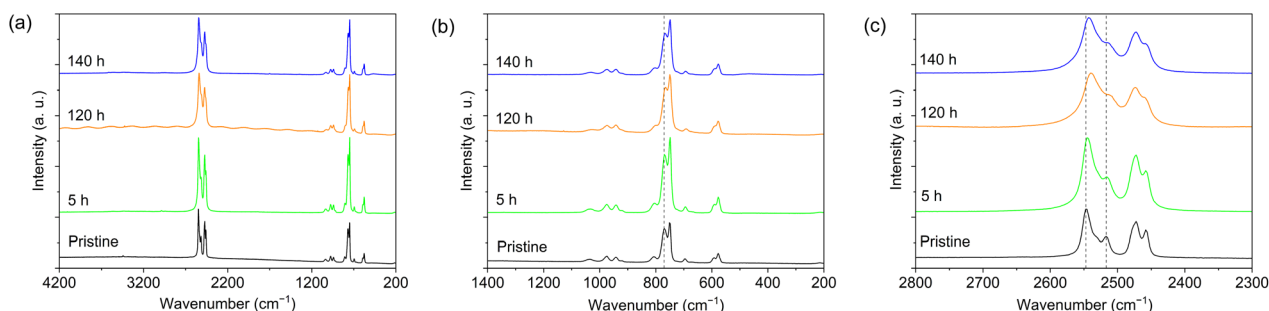


Fig. 2 Raman spectra of ball-milled samples with milling durations of 5, 120 and 140 h in comparison with pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$. (a) 4200–200 cm^{-1} , (b) 1400–200 cm^{-1} and (c) 2800–2300 cm^{-1} .

paddle-wheel mechanism.^{39,40} Atomic deficiencies have also been reported to result from ball-milling, which are believed to improve the ionic conductivity of $\text{Li}_2\text{B}_{12}\text{H}_{12}$.¹⁸ It can be noticed that Li^+ conductivity increases with increasing milling time from 5 to 120 h but drops when the time was further increased to 140 h. The highest ionic conductivity is observed for the sample milled for 120 h ($\sigma_{\text{Li}} = 3.71 \times 10^{-6}$ and $7.47 \times 10^{-4} \text{ S cm}^{-1}$ at 30 and 165 °C, respectively). A similar trend has been reported for ball-milled $\text{Li}_2\text{B}_{12}\text{H}_{12}\text{-Al}_2\text{O}_3$ composites.¹¹ The authors showed gradual conversions of RT $\alpha\text{-Li}_2\text{B}_{12}\text{H}_{12}$ to highly conductive $\beta\text{-Li}_2\text{B}_{12}\text{H}_{12}$ and H-deficient $\gamma\text{-Li}_2\text{B}_{12}\text{H}_{12}$ with a milling duration from 2–140 h by XRD. They proposed the conversion to $\beta\text{-Li}_2\text{B}_{12}\text{H}_{12}$ as the reason for high ionic conductivity in their 120 h milled $\text{Li}_2\text{B}_{12}\text{H}_{12}\text{-Al}_2\text{O}_3$ (25%) sample. As no new crystalline polymorphs are formed in the present study from XRD results, it is more likely that in this case, the extended ball-milling of 140 h may cause excessive amorphisation or disorder leading to the loss of some Li^+ migration channels and as a result, decreasing conductivity.

Partial thermal decomposition. The thermal behaviour of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ has been previously investigated and shows complex behaviour during heating, including desolvation, polymorphic phase transitions, and hydrogen release at high temperature, resulting in substoichiometric $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ compositions.³⁵ The reported $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ compounds display structural disorder, which may be of interest for the purposes of improved ion conductivity. As such, three samples were prepared by partially decomposing $\text{Li}_2\text{B}_{12}\text{H}_{12}$ at elevated temperature (380, 450, and 610 °C) under vacuum to synthesise $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ compositions with different stoichiometries. XRD patterns of the prepared $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples are shown in Fig. 4. A two-phase pattern is observed for the sample synthesised at 380 °C: sharp Bragg reflections at $d = 5.54$ and $4.79 \text{ \AA}$ from a minor proportion of undecomposed $\alpha\text{-Li}_2\text{B}_{12}\text{H}_{12}$ and a major nanocrystalline or amorphous diffraction halo at $d = 6.33 \text{ \AA}$ from H-poor $\gamma\text{-Li}_2\text{B}_{12}\text{H}_{12-y}$.³⁵ By conducting XRD with a known loading of crystalline internal standard (Y_2O_3 , Fig. S5†), the fractions of crystalline $\alpha\text{-Li}_2\text{B}_{12}\text{H}_{12}$ and amorphous/nano-crystalline $\gamma\text{-Li}_2\text{B}_{12}\text{H}_{12-y}$ are quantified as 7.4 and 92.6%, respectively. With higher decomposition temperatures (450 and 600 °C), $\alpha\text{-Li}_2\text{B}_{12}\text{H}_{12}$ is completely depleted and only a broad amorphous halo from $\gamma\text{-Li}_2\text{B}_{12}\text{H}_{12-y}$ is presented. The halo is observed at $d = 6.70 \text{ \AA}$ for the sample heated to 450 °C and shifts to much higher d -spacing for the sample heated to 610 °C ($d = 7.73 \text{ \AA}$). The movement of the $\gamma\text{-Li}_2\text{B}_{12}\text{H}_{12-y}$ halo to higher d -spacing is consistent with the steady removal of H at increasing decomposition temperature, which can lead to structural disorder and an increase in the average distance between ions in the solid. Formation of either LiH or amorphous boron⁴¹ is not observed in any samples.

The bonding environment in $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples was investigated by Raman spectroscopy (Fig. S6(a)–(c)†). Thermally decomposed $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples exhibited significant fluorescence, which became more significant in samples decomposed at higher temperature. Photoluminescence has been previously reported from $\text{Li}_2\text{B}_{12}\text{H}_{12}$ but has been commonly reported in derivatives with non-stoichiometric B:H ratios.^{42,43} Despite the fluorescence, vibrations for B–H bonds associated with $\text{B}_{12}\text{H}_{12}^{2-}$ cages are observed for the 380 °C sample with the modes at ≈ 2550 and 770 cm^{-1} being slightly shifted to lower wavenumbers compared with pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$ (see Fig. S6(b) and (c)†), indicating a disordered structure.^{13,18,36}

The Raman modes of $\text{B}_{12}\text{H}_{12}^{2-}$ are not well resolved in the spectrum for the sample heated to 450 °C, with two weak and broad bands being observed in the B–H stretching and bending regions between 700 and 1000 cm^{-1} and 2300 and 2700 cm^{-1} , respectively. For the sample heated to 610 °C, no Raman active signals are detected. To avoid the fluorescence effects from laser excitation, $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples were further studied by FTIR spectroscopy (Fig. 5). Bands at 1616 cm^{-1} and $3400\text{--}3800 \text{ cm}^{-1}$ are ascribed to O–H bending and stretching modes due to atmospheric moisture during data collection.⁴⁴ Typical signals of $\text{B}_{12}\text{H}_{12}^{2-}$ cages are the B–H bending mode (δ_{BH}) at $\approx 715 \text{ cm}^{-1}$, B–B stretching mode (ν_{BB}) at 1074 cm^{-1} , and B–H stretching modes (ν_{BH} , symmetric and asymmetric vibrations) appearing as broad bands at $\approx 2500 \text{ cm}^{-1}$.^{44–46} In the spectra of $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples, ν_{BB} and δ_{BH} modes did not shift from 1074 and 716 cm^{-1} , suggesting that the samples retain the icosahedral B_{12} skeletons, despite some broadening of the bands. The band of the ν_{BH} modes is observed at 2457 cm^{-1} for the sample

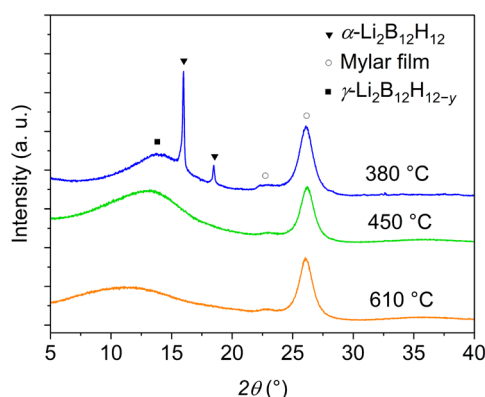


Fig. 4 XRD patterns of $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples synthesised by partial thermal decomposition at 380, 450 and 610 °C *in vacuo*.

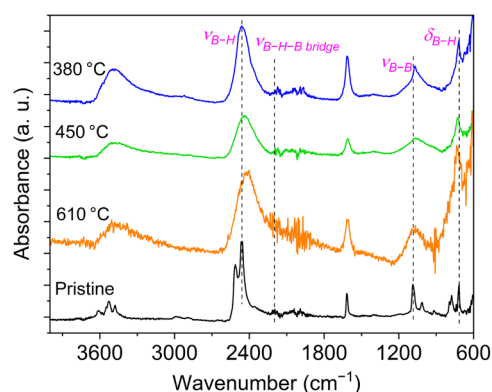


Fig. 5 FTIR spectra of $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples after partial thermal decomposition at 380, 450 and 610 °C *in vacuo*.

heated to 380 °C. This band red-shifts for analogues synthesised at higher temperatures, implying the weakening of B–H bonds.

To the best of our knowledge, the exact structures of boron–hydrogen species from thermal decomposition of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ remain unknown. Solid-state magic angle spinning nuclear magnetic resonance (MAS NMR) spectroscopy studies on the thermal decomposition of anhydrous alkali metal *closo*-dodecahydrododecaborates ($\text{M}_2\text{B}_{12}\text{H}_{12}$) showed a downfield shift of ^{11}B resonance after decomposition above 450 °C and it was proposed that this was due to a two-step decomposition mechanism involving the polymerization of $\text{M}_2\text{B}_{12}\text{H}_{12-y}$ to $(\text{M}_2\text{B}_{12}\text{H}_2)_n$.^{47,48} By XRD, Pitt *et al.*³⁵ showed that the *Pa*3 crystal structure of $\alpha\text{-Li}_2\text{B}_{12}\text{H}_{12}$ is irreversibly lost after dehydrogenation when attempting rehydrogenation of H-deficient $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ polymorphs. These authors argued that $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ phases exist in nanocrystalline form, rather than amorphous form and isolated H-deficient $\text{B}_{12}\text{H}_{12-y}$ icosahedra are retained in an expanded unit cell, based on the lack of a band around 2200 cm^{-1} for B–H–B linkages in known polymeric structures of $\text{B}_{12}\text{H}_{12}^{2-}$ ($\text{B}_{24}\text{H}_{23}^{3-}$, $\text{B}_{36}\text{H}_{34}^{4-}$ and $\text{B}_{48}\text{H}_{45}^{5-}$).^{49–54} Although we also do not observe a clear B–H–B band, it is prudent not to completely exclude the polymerization of B_{12} cages in $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples. Firstly, there is a shoulder at $\approx 2200 \text{ cm}^{-1}$ for the B–H–B bridging region in the FTIR data for our 610 °C sample, which may indicate some minor polymerisation. Secondly, the aggregation of B_{12} cages may proceed *via* other linking bonds, such as B–B and double B–H–B bonds,^{55–57} which are FTIR inactive. Good examples are isomers of $\text{B}_{20}\text{H}_{18}^{2-}$ from the oxidation of $\text{B}_{10}\text{H}_{10}^{2-}$: *iso*- $\text{B}_{20}\text{H}_{18}^{2-}$, composed of two $\text{B}_{10}\text{H}_9^{2-}$ fragments joined by two B–H–B bonds, shows IR ν_{BH} around 2500 cm^{-1} and ν_{BHB} at 1773 cm^{-1} , and *trans*- $\text{B}_{20}\text{H}_{18}^{2-}$, composed of two $\text{B}_{10}\text{H}_9^{2-}$ fragments joined by a B–B bond, shows IR ν_{BH} at 2525, 2493 cm^{-1} .^{57,58} In summary, based on results in the present study, it is difficult to assign specific structures for thermally decomposed $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples although it is expected that $(\text{Li}_2\text{B}_{12}\text{H}_2)_n$ polymers, perhaps $\text{Li}_2\text{B}_{12}\text{H}_{10}$ compositions,⁵⁹ or even mixtures of these species have been formed. Further studies, including theoretical calculations and simulations, may help identifying the structures.

Fig. 6 provides ionic conductivity results of the prepared $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples. The material partially decomposed at 380 °C exhibits $\sigma_{\text{Li}} = 1.03 \times 10^{-6} \text{ S cm}^{-1}$ at 30 °C, which is ≈ 100 times higher than that for pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$ ($\sigma_{\text{Li}} = 1.74 \times 10^{-8} \text{ S cm}^{-1}$). For the same sample, upon heating to 165 °C, a σ_{Li} of $1.30 \times 10^{-4} \text{ S cm}^{-1}$ was obtained. Meanwhile, no significant improvement was observed for the $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ analogues synthesised at higher temperatures, indicating that there is an optimal level of disorder in promoting Li^+ ion migration. In Nyquist plots (Fig. S7†), significantly lower resistance, R_{total} (sum of bulk and grain or phase boundary resistance),⁶⁰ was observed for the 380 °C sample compared with pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$ and other $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ analogues. Lower numbers of grain or phase boundaries are expected for all $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples because of their amorphous phase. Thus, the improved ionic conductivity only observed in the 380 °C sample is likely related to higher charge transfer within the bulk of the grains of

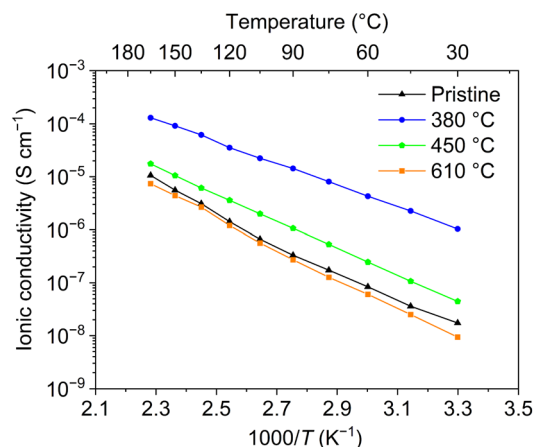


Fig. 6 Ionic conductivity of $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ samples as a function of temperature, measured isothermally by EIS. Decomposition temperatures of the samples are specified in the legends.

this electrolyte. Based on the Arrhenius equation $\sigma(T) = \sigma_0 \exp(-\Delta E_a/kT)$, the activation energy for cation diffusion (ΔE_a) for the 380 °C sample was found to be 39.5 kJ mol^{-1} , which is considerably lower than for pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$ ($\Delta E_a = 52.7 \text{ kJ mol}^{-1}$). ΔE_a values for the 450 and 610 °C samples are 48.8 and 55.0 kJ mol^{-1} , respectively, suggesting that it is harder for Li^+ to move in these samples. Some electrolytes exhibit higher ionic conductivity in an amorphous form, for instance Li_3PS_4 , because of mitigated grain boundary resistance.⁶¹ However, in some cases, such as $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, amorphisation can result in destruction of ion diffusion pathways leading to decreased conductivity.⁶² From these perspectives, we believe that the coexistence of crystallite and amorphous phases in the 380 °C sample may reduce bulk and grain boundaries while maintaining effective Li^+ transport pathways thus improving Li^+ conductivity.

It is of interest to compare the effects of partial thermal decomposition with mechanical ball-milling. Although no proportional relationships between Li^+ conductivity and disordering intensity (decomposition temperature and ball-milling duration) are found, the results here show that an improvement by approximately 2 orders of magnitude in low-temperature ionic conductivity by prolonged high-energy ball-milling of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ can be reproduced by facile thermal treatment. Based on structural data and the discussion above, expansion of the unit cell to accommodate H-deficient $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ after thermal treatment is expected and therefore a parallel exists with the crystalline lattice expansion observed after ball-milling. In addition, both ball-milling and thermal treatment can lead to formation of amorphous or nanocrystalline $\text{Li}_2\text{B}_{12}\text{H}_{12}$ with broad Bragg peaks occurring in the $\geq 5.8 \text{ \AA}$ *d*-spacing range, as previously reported for ball-milled $\text{Li}_2\text{B}_{12}\text{H}_{12}\text{-Al}_2\text{O}_3$ composites.⁴⁴ Therefore the same mechanism for cation migration may occur in ball-milled and thermal treated materials. It is noteworthy to mention that the sodium salts of the $\text{B}_{24}\text{H}_{23}^{3-}$ dimer and $\text{B}_{36}\text{H}_{34}^{4-}$ trimer anions have recently been reported to exhibit ionic conductivity of 2 orders of magnitude higher than

that of the monomer $\text{Na}_2\text{B}_{12}\text{H}_{12}$.^{49,50} These B–H–B bridged compounds may show similarities to the H-deficient $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ compounds in the present study.

Hydration. There are two known hydrated $\text{Li}_2\text{B}_{12}\text{H}_{12}$ crystal structures: $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 4\text{H}_2\text{O}$ and $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$, which show Li–Li distances of 3.332 and 3.005 Å, respectively compared to 5.232 Å in the pristine material's crystal structure.^{10,63,64} The known hydrates were prepared along with sub-stoichiometric hydrates ($\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ with $0 < x < 4$) to assess their electrochemical properties. A variation in the ratios of solvated molecules has been previously reported to affect ionic conductivity of metal hydridoborate compounds.^{31–33,65} XRD patterns of $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ samples are provided in Fig. 7 displaying mixtures of crystalline anhydrous $\text{Li}_2\text{B}_{12}\text{H}_{12}$, $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 4\text{H}_2\text{O}$ and $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$, in good agreement with data previously reported in the literature.^{35,63,64} $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ analogues with $0 < x < 4$ do not show new crystalline compounds, but only combinations of Bragg reflections from $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 4\text{H}_2\text{O}$ and pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$.

TGA curves of $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ samples with $0 < x \leq 4$ show mass loss over two events above 150 and 400 °C, respectively (Fig. S8†). The mass loss at lower temperature originates from dehydration, while the mass loss at higher temperature is attributed to the release of H_2 during partial decomposition of the remaining $\text{Li}_2\text{B}_{12}\text{H}_{12}$.^{35,64} This suggests that the samples remain hydrated up to at least 150 °C. Uniquely, TPPA results (Fig. S9(a)†) indicate that $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$ begins to melt at ≈ 115 °C and becomes a liquid at higher temperatures. However, this was only reliably observed in a sealed capillary, where water vapour is not able to be released due to dehydration (*i.e.* with a back pressure of H_2O). DSC-TGA performed with $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$ sealed in an Al crucible (Fig. S9(b)†) shows an endothermic phenomenon related to its melting point at 115 °C instead of two dehydration steps at 56 and 151 °C as reported in the literature when an open crucible was used.^{35,64} In addition, no mass loss is observed while heating the sample in a sealed Al crucible, in opposition to the dual step mass loss seen from dehydration in an open crucible. Based on these observations, it is concluded that under a closed system, such as inside an

electrochemical test cell, the solid $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$ will remain in a powder form up to ≈ 115 °C, before melting.

Ionic conductivity data for the $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ series are provided in Fig. 8. Compared with pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$, $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ samples with $0 < x \leq 4$ show no significant enhancement. On the other hand, $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$ exhibits an increase of approximately 1 order of magnitude in ionic conductivity compared to the lower hydrates and pristine material. It exhibits $\sigma_{\text{Li}} = 1.33 \times 10^{-7}$ and $5.10 \times 10^{-5} \text{ S cm}^{-1}$ at 30 and 105 °C, respectively. It should be noted that any beneficial effect from ball-milling in the synthesis of these samples should be negligible as a short milling duration of 10 min also results in no difference in the ionic conductivity of pristine $\text{Li}_2\text{B}_{12}\text{H}_{12}$ (Fig. S10†). Li^+ is the dominant charge carrier in $\text{Li}_2\text{B}_{12}\text{H}_{12}$ with Li^+ transference number > 0.9 .³⁶ The Li^+ transference number of $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$ was determined using polarisation responses in a chronoamperometry measurement³⁶ where a constant voltage of 50 mV was applied to a $\text{Li}|\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}|\text{Li}$ symmetric cell at 80 °C for 20 min (Fig. S11†). Identical currents were observed throughout the measurement, within uncertainty, and the Li^+ transference number was calculated to be ≈ 1 . Thus, the ionic conductivity in $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$ is dominated by Li^+ . $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$ has a layered structure, analogous to the orthorhombic structure of $\text{BaB}_{12}\text{H}_{12} \cdot 6\text{H}_2\text{O}$, with isolated $[\text{Li}_2(\text{H}_2\text{O})_7]^{2+}$ cation pairs being formed by the corner sharing of two $[\text{Li}(\text{H}_2\text{O})_4]^+$ as characteristic features.⁶⁴ The structure is stabilised by intermolecular interactions between $[\text{Li}_2(\text{H}_2\text{O})_7]^{2+}$ units and $\text{B}_{12}\text{H}_{12}^{2-}$ anions *via* $\text{B}-\text{H}^{\delta-} \cdots \delta^+\text{H}-\text{O}$ hydrogen bonds without direct coordination of Li^+ cations with the anions, in contrast to $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 4\text{H}_2\text{O}$ and $\text{Li}_2\text{B}_{12}\text{H}_{12}$ structures, where Li^+ cations are directly bonded to the H atoms of $\text{B}_{12}\text{H}_{12}^{2-}$ anions.^{10,63} From crystal data available in the literature, the Li–Li distances are determined to be slightly larger for $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$ than $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 4\text{H}_2\text{O}$, at 3.332 and 3.005 Å, respectively.^{63,64} As such, it is proposed that the weaker cation–anion interactions in $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot 7\text{H}_2\text{O}$ may

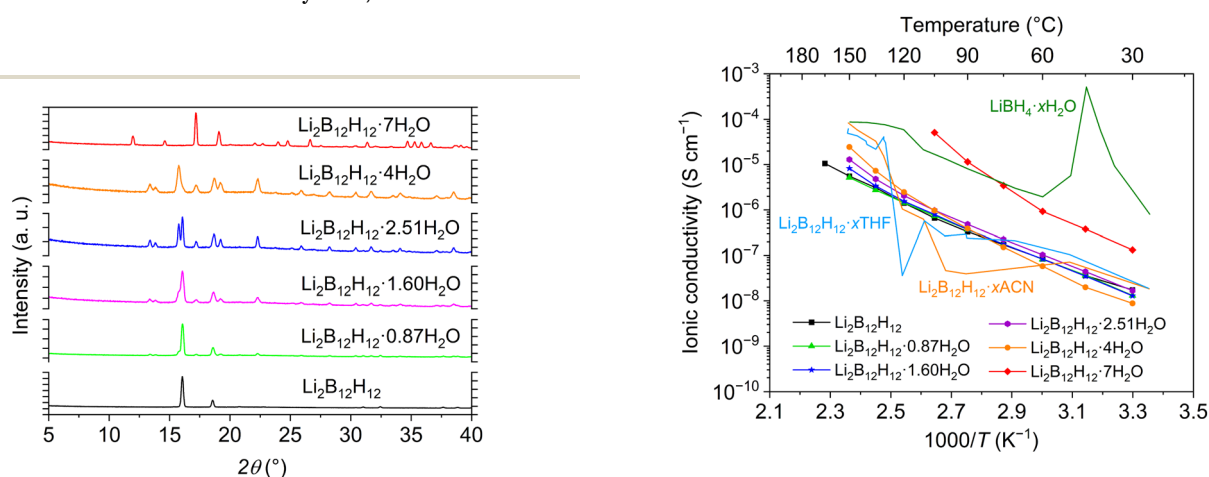


Fig. 7 XRD patterns of $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ samples with $x = 0-7$. The amount of coordinated water for $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ samples with $0 < x \leq 4$ was calculated from TGA data in Fig. S8†.

Fig. 8 Ionic conductivity of $\text{Li}_2\text{B}_{12}\text{H}_{12} \cdot x\text{H}_2\text{O}$ samples ($x = 0-7$) as a function of temperature, measured isothermally by EIS (filled symbols) compared to solvated Li-hydridoborates in the literature (lines).^{17,26}

facilitate different reorientational dynamics of $B_{12}H_{12}^{2-}$ and enhanced Li^+ mobility.

The effect of neutral solvent ligands has been reported for some metal hydridoborates.^{26–29,31–33} Kisu *et al.*²⁹ reported Zn^{2+} and Mg^{2+} conductivity on the order of $10^{-5} S cm^{-1}$ at 50 °C for $ZnB_{12}H_{12} \cdot 12H_2O$ and $MgB_{12}H_{12} \cdot 12H_2O$ and proposed a rapid exchange between coordinated and non-coordinated crystal water as one of the mechanisms for cation conduction in the compounds based on nuclear magnetic resonance spectroscopy results. Yan *et al.* studied Mg^{2+} conduction in $Mg(BH_4)_2 \cdot xNH_3$ and reported a Mg^{2+} conductivity of $3.3 \times 10^{-4} S cm^{-1}$ at 80 °C for $Mg(BH_4)_2 \cdot NH_3$, which is ≈ 8 orders of magnitude higher than the value for $Mg(BH_4)_2$ at the same temperature.³¹ $Mg(BH_4)_2 \cdot NH_3$ was shown to have an orthorhombic crystal structure constructed of $-BH_4-Mg-BH_4-Mg-$ chains interlinked by $N-H^{\delta+} \cdots \delta^-H-B$ dihydrogen bonds between H atoms from NH_3 ligands and BH_4 anions. Based on density functional theory (DFT) simulations, the authors proposed the transfer of NH_3 molecules among lattice Mg^{2+} and interstitial Mg^{2+} in a manner similar to a *pas de deux* as the mechanism for Mg^{2+} conduction in the ammine magnesium borohydride complex.³¹ One may expect a similar mechanism for $Li_2B_{12}H_{12} \cdot 7H_2O$ because of its structural similarities to $Mg(BH_4)_2 \cdot NH_3$. Theoretical studies may also need to be employed to confirm the mechanism of Li^+ conduction in $Li_2B_{12}H_{12} \cdot 7H_2O$.

Electrochemical stability

Electrochemical oxidative stability of pristine $Li_2B_{12}H_{12}$, $Li_2B_{12}H_{12-y}$ synthesised at 380 °C, 120 h ball-milled $Li_2B_{12}H_{12}$, and $Li_2B_{12}H_{12} \cdot 4H_2O$ was investigated by LSV from $-0.4 V$ lower than OCP to $+6 V$ (vs. Li^+/Li) with a slow scan rate of $0.05 mV s^{-1}$ (Fig. 9). The oxidation of pristine $Li_2B_{12}H_{12}$ is observed with a minor event at $+3.2 V$ and a major event at $+3.6 V$, in good agreement with the literature.^{11,66} Surprisingly, $Li_2B_{12}H_{12-y}$ synthesised at 380 °C, over 3 repetitive measurements with a fresh sample, shows a flat current profile with no significant anodic current throughout the swept potentials, even at

a voltage greater than $+6 V$. On the other hand, the ball-milled and hydrated samples show aggressive oxidation at $\approx +2.93$ and $+3.75 V$, respectively. This contrasts with results reported by Kim *et al.*, which proposed an electrochemical stability window of -0.1 to $+5 V$ for a ball-milled $Li_2B_{12}H_{12}$ sample.¹⁸ However, these authors may have overestimated the stability of their sample since a fast scan rate of $5 mV s^{-1}$ was used in cyclic voltammetry.¹² It is known that the modest oxidative stability limits of the *closo*- $B_{12}H_{12}^{2-}$ anion, and other metal hydridoborates, result from the electrochemical oxidation of these compounds through the loss of hydrogen, such as in thermal decomposition. The electrochemical oxidation of $B_{12}H_{12}^{2-}$ in acetonitrile at a Pt electrode is found to occur at $+1.5$ and $+1.85 V$ (vs. SCE) via the dimerisation of two anion cages into the $B_{24}H_{23}^{3-}$ dimer.^{53,54} Nevertheless, oxidation products of $B_{12}H_{12}^{2-}$ have been recently reported to show high electrochemical stability. Stepwise cyclic voltammetry with a $Na_4(-CB_{11}H_{12})(B_{12}H_{12})$ composite electrolyte shows decomposition of both anions in the first forward scan to $+5 V$ vs. Na^+/Na but only low anodic currents are observed in the second forward scan, demonstrating that no further electrochemical oxidation of the decomposition products occurs.⁶⁶ Sodium salts of dimer and trimer anions from thermal protolysis of $B_{12}H_{12}^{2-}$, namely $Na_3B_{24}H_{23}$ and $Na_4B_{36}H_{34}$, have been reported to exhibit oxidative stability limits above $+6 V$ vs. Na^+/Na .^{49,50} Based on these points, the excellent oxidative stability observed for $Li_2B_{12}H_{12-y}$ synthesised at 380 °C may be explained by the fact that the SSE, different from the ball-milled and hydrated samples, already exists in an oxidised state.

The electrochemical stability window of the $Li_2B_{12}H_{12-y}$ electrolyte synthesised at 380 °C was further evaluated by a LSV measurement sweeping from an OCP down to $1 V$ (vs. Li^+/Li). The sample shows a cathodic peak at $\approx 1.92 V$ (vs. Li^+/Li) while no obvious reaction is observed for a pristine $Li_2B_{12}H_{12}$ sample (Fig. S12†). The peak may be related to the electrochemical reduction of oxidised species formed from the partial thermal decomposition of $Li_2B_{12}H_{12}$. This result is interesting in that the

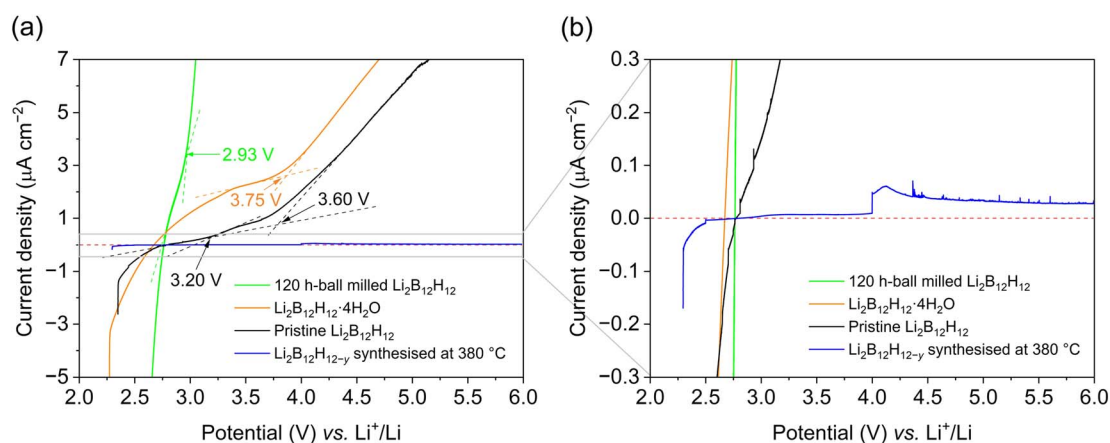


Fig. 9 (a) Electrochemical oxidative stability (vs. Li^+/Li) measured at 80 °C for 120 h-ball milled $Li_2B_{12}H_{12}$, $Li_2B_{12}H_{12} \cdot 4H_2O$, pristine $Li_2B_{12}H_{12}$ and $Li_2B_{12}H_{12-y}$ synthesised by partial thermal decomposition at 380 °C, and (b) zoomed-in plot for the $Li_2B_{12}H_{12-y}$ synthesised by partial thermal decomposition at 380 °C.

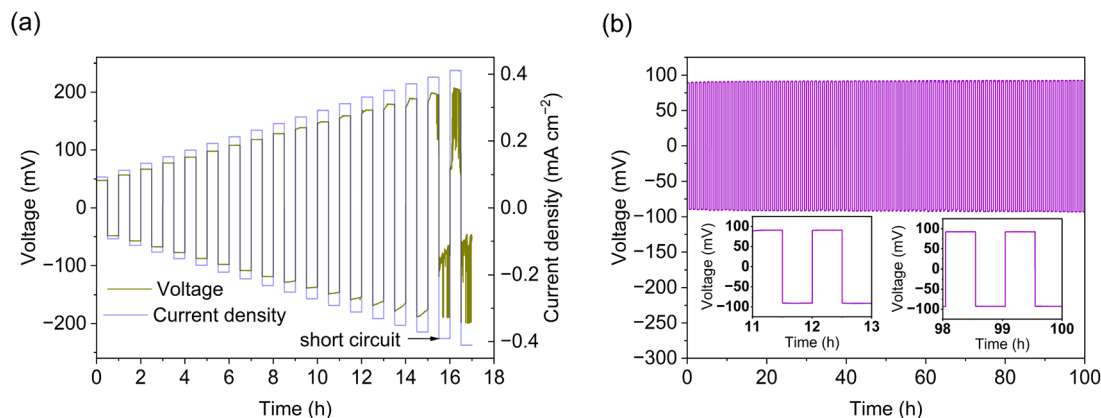


Fig. 10 (a) Critical current density by stepwise GCD (vs. Li^+/Li) with $\text{Li}|\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ synthesised at $380\text{ }^\circ\text{C}|\text{Li}$ symmetric cell and (b) GCD with the symmetric cell at a current density of 0.17 mA cm^{-2} for 100 h at $130\text{ }^\circ\text{C}$.

material shows excellent oxidative stability, but some potential limitations in its reductive stability.

Electrochemical oxidative stability is crucial to high voltage batteries but for battery cycling the compatibility of the SSE with the anode side is also important. Therefore, the electrochemical reductive stability of $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ synthesised at $380\text{ }^\circ\text{C}$ towards a Li metal anode *via* GCD measurements with a $\text{Li}|\text{SSE}|\text{Li}$ symmetric system was evaluated. The critical current density, where short circuits occur due to Li dendrite growth, of the SSE is found at $\approx 0.39\text{ mA cm}^{-2}$ at $130\text{ }^\circ\text{C}$ (Fig. 10(a)), which is higher than values found for pristine and 120 h ball-milled $\text{Li}_2\text{B}_{12}\text{H}_{12}$ (0.23 and 0.11 mA cm^{-2} , respectively (Fig. S13†)). This suggests that the $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ sample has an ability to deliver more stable Li plating-stripping compared with the pristine and ball-milled samples. Furthermore, in a long-term GCD measurement at 0.17 mA cm^{-2} at $130\text{ }^\circ\text{C}$ (Fig. 10(b)), the symmetric cell of this SSE exhibits a flat voltage-time profile at each cycle (as shown in the inset figures) with no obvious variation of overpotentials for the Li plating-stripping voltage profile over 100 h of cycling (equivalent to 100 of the GCD cycles). Consequently, the results here strongly indicate that the SSE synthesised by partial thermal decomposition of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ at $380\text{ }^\circ\text{C}$ has good Li dendrite suppression capacity and can form stable interfaces with the Li metal anode.

Conclusions

A $\text{Li}_2\text{B}_{12}\text{H}_{12}$ solid electrolyte has been modified by mechanical ball-milling, partial thermal decomposition and solvation with H_2O ligands. Ball-milling is shown to introduce strain and/or defects in the $\text{Li}_2\text{B}_{12}\text{H}_{12}$ lattice, improving ionic conductivity. Partial thermal decomposition of $\text{Li}_2\text{B}_{12}\text{H}_{12}$ at $380\text{ }^\circ\text{C}$ results in an increase in Li^+ conductivity by ≈ 2 orders of magnitude but higher decomposition temperatures lead to no notable improvement. For the hydrated samples, enhanced ionic conductivity is observed for a sample with 7 crystal H_2O molecules. In terms of electrochemical stability, the SSE synthesised by partial thermal decomposition at $380\text{ }^\circ\text{C}$ is found to be stable even at potentials greater than $+6\text{ V}$ (vs. Li^+/Li) whilst ball-

milling and hydrated samples have oxidative stability limits below $+4\text{ V}$. This work proposes partial thermal decomposition as a novel pathway to improve the ionic conductivity and electrochemical stability of metal hydridoborate-based solid electrolytes, offering a new tool for tuning the properties of these materials. Charge-discharge cycling with battery cells of the $\text{Li}_2\text{B}_{12}\text{H}_{12-y}$ electrolyte synthesised at $380\text{ }^\circ\text{C}$ could not be demonstrated due to its poor mechanical strength. The partially thermally decomposed sample is brittle and ceramic-like, which hinders the formation of stable interfaces with a cathode. Electrode-electrolyte interface engineering strategies to improve interfacial contacts are crucial to employ the material in a full ASSB system.

Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

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